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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

TU856 - BSc. (Honours) Degree in Computer Science

Year 3

SEMESTER 1 EXAMINATIONS 2024/25

CMPU3038 - Software Engineering III

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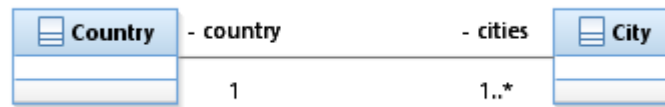
Exam Duration: 2 hours

Instructions: e.g. Answer three out of four questions. All questions carry 33 marks each. One complimentary mark shall be awarded.

Special Instructions/Handouts:

If asked in any question to provide an example of code, you may use any appropriate language of your choice or pseudo code in your answer.

Q.1 (a) Given the structural model below. Answer the following questions.



- (i) Explain, in detail, the meaning of the association between the two classes. (5 marks)
 - (ii) Using the example, explain what is meant by *referential integrity* in the context of Object-Oriented Programming. (5 marks)
 - (iii) Show, using code examples, how referential integrity would be implemented for this example. (12 marks)
- (b) The *Principle of Least Knowledge* promotes decoupling so that communication only occurs between appropriate objects. Using a suitable class diagram and sequence diagram example, explain how this principle can be breached while still allowing correct business logic to be performed. (11 marks)

Q.2 (a) (i) Briefly describe what is meant by a *structural* software design pattern. (4 marks)

- (ii) There are a number of similar *structural design patterns* that are often grouped together under the name *wrapper* patterns. Identify **three** of these patterns giving their *intent* and provide a UML diagram that describes their structure. In your answer, ensure you identify why they are considered as *wrapper* patterns. (6 marks x 3)
- (b) The *intent* of the *Model View Controller* design pattern is to "*separate the application logic from the user interface logic*". In the context of a web application where the presentation is generated on the server side and passed to the client, explain, with the aid of a clearly labelled diagram, how this design pattern could be realised. (11 marks)

Q.3 The *Data Access Object (DAO)* is a well-known and popular design pattern. Answer each of the following questions in relation to the DAO:

- (a) Discuss its intent. (6 marks)
- (b) Draw a UML structural diagram that would illustrate the organisation and associations of classes collaborating in the DAO pattern. (9 marks)
- (c) In terms of *abstraction*, discuss the benefits it provides. (9 marks)
- (d) Provide a code example illustrating its implementation and use. (9 marks)

Q.4 (a) (i) Outline what is meant by *White Box* testing and *Black Box* testing. In your answer, ensure you clearly identify how they differ. (5 marks)

(ii) In the context of developing Object-Oriented systems, briefly describe what is meant by *scenario-based* test design. (5 marks)

(b) (i) Describe what is meant by *Unit Testing*. In your answer ensure you provide the goal and benefits of this testing approach. (6 marks)

(ii) Explain what is meant by *Test Driven Development* and show how it differs from a "traditional" approach to *Unit Testing*. (9 marks)

(iii) Discuss the role of *Mock Objects* when implementing unit tests. (8 marks)